

EXHAUSTIVE ACADEMIC REVIEW PROTOCOL

**ELECTRICAL INSTRUMENTATION: PHASE 6**

**Exam Hall Protocol & Time Management**

*The 3-Hour Execution Strategy and Quality Control Checklist*

Semester Preparation Masterclass

Time Remaining: < 48 Hours

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## Tackling Question 1: "Justify or Correct"

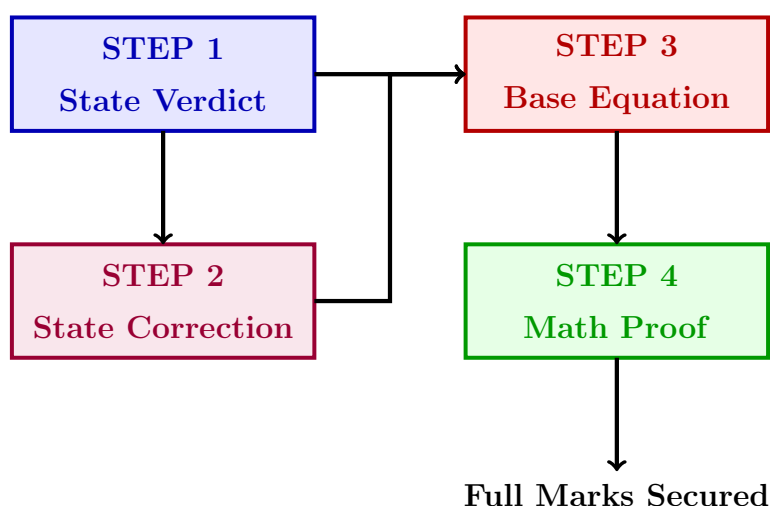
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Question 1 is almost always mandatory and consists of 4 to 5 "Justify or Correct" statements. This is where BJB Sir tests if your understanding is surface-level or grassroots. Do not write essays. Write mathematical proofs.

### The 4-Step Answering Algorithm

Every answer in Q1 must follow this exact structural algorithm to secure full marks:

1. **The Absolute Verdict:** Start the answer by explicitly stating "**The statement is Correct**" or "**The statement is Incorrect.**" Do not be ambiguous.
2. **The Correction (If Applicable):** If incorrect, rewrite the statement correctly in one sentence. (e.g., "The sensitivity is inversely proportional to the square of the distance, not directly proportional.")
3. **The Grassroots Equation:** Write down the fundamental physics equation governing the system ( $C = \epsilon A/d$ ,  $q = dF$ , etc.).
4. **The Derivation/Proof:** Differentiate or manipulate the equation to prove your verdict. Let the math do the talking.



# The 180-Minute Time Management Matrix

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Assuming a 50-mark Part I paper requiring you to answer Question 1 (20 marks) and two other long-form questions (15 marks each).

## Recommended Time Split

- **First 10 Minutes (The Scan):** Do not write anything. Read the entire paper. Identify the exact derivations you know best. Choose your two long-form questions immediately.
- **Minutes 10 to 60 (Question 1):** Dedicate a full 50 minutes to the mandatory "Justify or Correct" section. Since there are 4-5 parts, spend exactly 10-12 minutes per part. Draw small, neat orthogonal diagrams even if not explicitly asked.
- **Minutes 60 to 110 (Long Question A):** Spend 50 minutes executing your strongest long-form question. This is where you lay down the massive grassroots derivations (e.g., Push-Pull Linearization, Charge Amplifier Virtual Ground).
- **Minutes 110 to 160 (Long Question B):** Spend 50 minutes on the final long-form question or the numericals. Show all units and intermediate steps in calculations.
- **Minutes 160 to 180 (Quality Control):** The final 20 minutes are for the BJB Sir Submission Checklist (see Section 3).

# The "BJB Sir" Quality Control Checklist

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Before you hand in your answer script, use your final 20 minutes to scan through your answers and verify the following criteria. BJB Sir deducts marks for missing these engineering details.

## 1. Diagrammatic Integrity

- Did I use a ruler for every single block, wire, and dimension line?
- Are my dimension lines (like gap  $d$  or thickness  $t$ ) strictly orthogonal to the plates?
- Did I ensure absolutely **zero text overlaps** with any geometric lines?
- Did I label the specific materials? (e.g., "Nickel-Iron Core", "Teflon Coating", "Mu-metal Sheath").

## 2. Derivation Rigor

- Did I explicitly write down the **Assumptions**? (e.g., "Assuming fringing fields are negligible due to guard rings," or "Assuming displacement  $x \ll d$ ").
- Did I define every variable I introduced? (e.g., "Where  $\epsilon_r$  is the relative permittivity of the solid specimen").
- Did I conclude the derivation with a sentence explaining the **physical engineering significance**? (e.g., "This proves the non-linearity is annihilated").

## 3. System Awareness

- For any LVDT question, did I mention the **Longitudinal Slot** in the core?
- For any piezoelectric question, did I mention the **Charge Amplifier** and the **Virtual Ground**?
- For capacitive thickness/variable-gap questions, did I mention **Fringing**?
- Did I include the correct units at the end of every numerical answer? (pF, m/s, Volts,  $\mu\text{V}$ ).

You have the blueprints. You have the derivations.

You have the strategy.

Trust the physics, keep your lines straight, and  
execute the math.

**Good Luck on the Semester Examination.**